

A Specific View of Global Trade Shocks

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Abstract

We develop simple sufficient statistics for the short and long-run gains from trade in a multi-industry Armington model with industry-specific capital, arbitrary country-by-industry differences in factor and input intensities, and circular input-output linkages. Starting from a long-run equilibrium, the change in the price index from a return to autarky decomposes into three terms: the standard gains from trade formula with inputs, a factor proportions term that depends only on the share of capital in value-added before and after the shock, and a misallocation term that captures the cost of capital being stuck in its industry. The misallocation term is a function of the change in each industry's share of total capital payments, and can therefore be quantified using national accounts data by industry, without any trade data or trade elasticities. Using the World Input-Output Database, we find that specific factors raise the cost of autarky only modestly in the aggregate, adding 3.7% to prices on average across 43 countries in 2014. However, this masks large differences across capital owners: in the US, capital owners lose 4% in the long run, while owners of textile capital gain 158% in the short run and owners of machinery capital lose 46%. Since 1965, misallocation has become less important relative to the standard gains from trade, and industry-specific returns to autarky have shifted substantially, often in line with known shocks such as the phase-out of textile quotas.

JEL Codes: F11, F13, F14, F16

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1 Introduction

The political economy of globalization often intersects with capital ownership. In the classic analysis of Mayer (1984), inequality in the ownership of factors determines the tariff policy preferred by the median voter, and thus the direction in which trade policy is likely to move. More generally, the aggregate gains from trade policy are rarely shared by all factor owners, and unless suitable transfers are made, the distribution of factor ownership shapes who supports and who opposes greater openness. The timing of factor reallocation matters as well. When capital is specific to an industry in the short run, and only mobile across industries in the long run, the gains and losses to factor owners can differ dramatically across time horizons. Indeed, long-run gains can be outweighed by near-term losses if agents are sufficiently impatient (Neary, 1978; Artuç et al., 2010).

These issues have become more pressing in recent years. While a return to autarky was once mostly an intellectual exercise used to benchmark the gains from trade, the recent wave of protectionist measures, and continued threats of more, have made large trade shocks, and their distributional consequences, a much more practical concern. In this paper, we ask two related questions. First, how do we characterize the difference between the short and long-run welfare effects of a trade shock when some factors are fixed within industries, and what are the implications for inequality across factor owners? Second, how large are these differences in practice, and how have they evolved over the last half-century?

To answer these questions, we develop an Armington model of trade in the style of Arkolakis et al. (2012, henceforth ACR) in which each industry uses specific factors (capital), labor, and intermediate inputs. The model has three key features. First, we allow for arbitrary differences in the intensity of the fixed factor by country and industry, which leads to country-by-industry differences in the elasticity of supply. Second, within the variable factors, we allow for arbitrary country-by-industry differences in labor and input intensity. Third, we allow for circular input-output linkages in the spirit of Eaton and Kortum (2002) and Alvarez and Lucas (2007). We then compare the short-run version of the model, in which capital cannot move across industries, with the long-run version, in which it can, and develop sufficient statistics for the move from one to the other.

On the theoretical side, we show that the standard ACR formula for exact changes in the price index is adjusted, in both the short and long run, by the exact change in rental rates. In the short run, the exact change in the rental rate within an industry is equal to the exact change in the value of output in that industry. In the long run, the exact change in the (common) rental rate can be calculated using information on the change in labor shares, weighted by labor intensity across industries. Our main theoretical result focuses

on the case in which a country begins in a long-run equilibrium and returns to autarky. Here, the change in prices decomposes into three multiplicative components: (i) the ACR formula for a one-factor economy with intermediate inputs, (ii) a factor proportions term that depends only on the share of capital payments in value-added at baseline and in autarky, and (iii) a misallocation term that arises only when capital is specific. The misallocation term is a geometric average of the exact change in each industry’s share of total capital payments, weighted by those same shares in autarky, and scaled by the autarky share of capital in value-added. We show that this term is always weakly greater than one, and that it has a simple interpretation as the exponential of an information-theoretic distance (the Kullback-Leibler divergence) between the distribution of capital payments across industries at baseline and in autarky. An important implication is that quantifying the misallocation from specific factors only requires national accounts data by industry. No trade data, and no trade elasticities, are needed. This makes it straightforward to measure misallocation, and the returns to different factor owners, over a long period of time.

On the quantitative side, we calibrate the model to the World Input-Output Database (WIOD). We first evaluate the welfare effects of a return to autarky starting from 2014. In most countries, the aggregate misallocation from fixed factors is fairly small relative to the ACR component. In the US, prices rise 5.8% in the long run when moving to autarky, and an additional 0.6% in the short run. China is a notable exception. Prices in China rise 6.6% in the long run, and an additional 2.6% in the short run, so that misallocation raises the total price increase from autarky by roughly 40%. Across all 43 countries, the short-run price increase is 3.7% larger than the long-run price increase, on average.

The modest aggregate costs of specific factors mask substantial heterogeneity across capital owners in the short run. In the US, owners of capital in textiles, leather and footwear experience a 158% increase in real income when moving to autarky under the specific factors assumption, whereas the long-run real income of capital owners falls by 4%. At the other end of the distribution, owners of capital in machinery and transport equipment lose 46% and 31% of real income, respectively.

We then use the long-run version of WIOD to measure how the returns to autarky have changed from 1965 to 2014. We find dramatic changes in industry-specific returns to autarky over this period, with some industries switching from “anti-autarky” to “pro-autarky” under specific factors, and vice versa. For example, the short-run gain from autarky for capital owners in US textiles was comparatively modest, below 50%, through the early 1990s, and then rose sharply following the implementation of the Agreement on Textiles and Clothing and the removal of the remaining textile quotas in 2005. In contrast, capital owners in US electrical and optical equipment moved from substantial losses from autarky in the 1960s and

1970s to roughly zero by 2014. In terms of aggregates, the misallocation term has been fairly constant over time in levels, but its importance relative to the standard gains from trade has fallen as the gains from variety have risen with deeper trade integration. For China, prior to 1980, misallocation through specific factors was roughly as important as the traditional ACR calculation.

Related Literature Our paper is most closely related to the literature that quantifies the gains from trade using sufficient statistics, beginning with [Arkolakis et al. \(2012\)](#) and summarized in [Costinot and Rodríguez-Clare \(2014\)](#). Our long-run results are closely related to the extension with multiple factors and intermediate inputs discussed in the appendix of [Arkolakis et al. \(2012\)](#), and to input-output extensions such as [Caliendo and Parro \(2015\)](#). Relative to this work, we show that the long-run adjustment to the gains from trade formula with multiple factors can be summarized with a single statistic, the share of capital in value-added, and we characterize the additional costs that arise when capital is industry-specific.

The closest paper to ours is [Galle et al. \(2023\)](#), who embed an extended Roy model matching worker groups to industries within a quantitative trade model. They find significant heterogeneity in the welfare response to the China shock and to a move to autarky, with the gains from trade ranging from -4.82% to 3.03% across groups. Our focus is complementary. We study the distributional consequences of trade across owners of industry-specific capital, rather than across worker groups, and we emphasize the difference between short and long-run outcomes rather than a single equilibrium. Our framework is also simpler, in that the key sufficient statistics for misallocation can be computed from national accounts alone.

Our paper also relates to quantitative trade models with specific factors, diminishing returns, or external economies, including [Spearot \(2012, 2013\)](#), [Ruhl and Willis \(2017\)](#), [Anderson and Yotov \(2020\)](#), and [Farrokhi and Soderbery \(2024\)](#). In these models, an upward-sloping industry supply curve, whether due to fixed factors, capacity, or adjustment frictions, changes how trade shocks pass through to prices and factor returns. We share the emphasis on the short-run supply side, and focus on its implications for the gains from trade and factor returns. More broadly, work on the labor market adjustment to trade shocks, such as [Artuç et al. \(2010\)](#), [Kovak \(2013\)](#), [Dix-Carneiro \(2014\)](#), and [Caliendo et al. \(2019\)](#), emphasizes that factors are slow to reallocate, and that the transition between short and long-run equilibria matters for welfare.

Finally, there is the entire classical literature on factor endowments and specific factors, including [Jones \(1971\)](#), [Mayer \(1974\)](#), [Mussa \(1974\)](#), and [Neary \(1978\)](#), who study the short and long-run effects of trade and trade policy on factor returns. The main difference in our setup is that the Armington structure yields variety-style gains from trade and guarantees

complete diversification. As a consequence, love of variety pushes all factors toward losing in real terms when moving to autarky, even as relative factor returns respond to changes in the pattern of production in the familiar Stolper-Samuelson fashion. Indeed, in our quantification, both labor and capital lose in the long run in every country except Norway, where labor gains slightly.

The rest of the paper proceeds as follows. Section 2 outlines the model. Section 3 derives exact changes in prices in the short and long run. Section 4 studies a return to autarky and derives our main decomposition. Section 5 describes the data and calibration. Section 6 presents results for 2014, and Section 7 presents results from 1965 to 2014. Section 8 concludes and discusses ongoing work on the transition from the short to the long run.

2 Model

2.1 Consumers

Consumers in country l earn utility across I industries, and source products from J suppliers. We adopt the Armington assumption, where each country produces a specific variety within each industry, and consumers have CES demand across these varieties. Specifically, utility is written as:

$$U_l = \prod_{i \in I} u_{il}^{\theta_{il}} \quad (1)$$

where $\sum_i \theta_{il} = 1$, and sub-utility within each industry sold in country l is:

$$u_{il} = \left(\sum_{j \in J} q_{ijl}^{\frac{\sigma_{il}-1}{\sigma_{il}}} \right)^{\frac{\sigma_{il}}{\sigma_{il}-1}} \quad (2)$$

Here, σ_{il} is the elasticity of substitution across varieties within industry i in country l , and q_{ijl} is the quantity of the variety from j consumed in l . The budget constraint faced by consumers in l is

$$\sum_{i \in I} \sum_{j \in J} p_{ijl} q_{ijl} \leq Y_l^c \quad (3)$$

where Y_l^c is the income consumers spread across varieties. Later, we will see that producers also source final goods as inputs, which contributes to aggregate demand. To compare our results most closely to ACR, we assume that there is no tariff revenue.

Solving the consumer's problem, the price index for industry i in country l is:

$$P_{il}^{1-\sigma_{il}} = \sum_{j \in J} p_{ijl}^{1-\sigma_{il}} \quad (4)$$

and the aggregate consumer price index is $P_l = \prod_i (P_{il}/\theta_{il})^{\theta_{il}}$. As we describe below, P_l will also be the price of inputs for producers in country l . Finally, expenditure shares on the variety from j within industry i in country l are written as:

$$\lambda_{ijl} = \frac{p_{ijl}^{1-\sigma_{il}}}{P_{il}^{1-\sigma_{il}}} \quad (5)$$

These expenditure shares will feature prominently in the gains from trade formula that we derive shortly.

2.2 Production

We adopt a flexible Cobb-Douglas production framework for industry i in country j , where countries and industries vary in their intensity of the three factors we consider: labor, materials, and capital. We assume that labor and materials can both be adjusted in the short run, while capital can only be reallocated across industries in the long run.

Industry i in country j has K_{ij} units of capital, with β_{ij} the factor share of capital, and $1 - \beta_{ij}$ the share of variable factors. Within variable factors, γ_{ij} is the expenditure share on labor and $1 - \gamma_{ij}$ the expenditure share on inputs. Labor is purchased at wage w_j , and inputs are purchased in the same shares as final goods, at price P_j , the consumer price index in j . These assumptions are for tractability, and to provide a clear link to available input-output tables for later calibration.

Overall, total variable costs of production for industry i in country j , given capital K_{ij} , are written as:

$$C_{ij}(Q_{ij}) = w_j^{\gamma_{ij}} P_j^{(1-\gamma_{ij})} \frac{Q_{ij}^{\frac{1}{1-\beta_{ij}}}}{K_{ij}^{\frac{\beta_{ij}}{1-\beta_{ij}}}} \quad (6)$$

where we have normalized constants to one for ease of notation. Total production, Q_{ij} , must cover consumption in all destinations, adjusted by the iceberg cost of trade $\tau_{jl} \geq 1$:

$$Q_{ij} = \sum_l \tau_{jl} q_{ijl} \quad (7)$$

Firms price competitively, with price equal to marginal cost by destination. Accordingly, the price in country l for goods from j is written as:

$$p_{ijl} = \frac{w_j^{\gamma_{ij}} P_j^{(1-\gamma_{ij})}}{1 - \beta_{ij}} \frac{Q_{ij}^{\frac{\beta_{ij}}{1-\beta_{ij}}}}{K_{ij}^{\frac{\beta_{ij}}{1-\beta_{ij}}}} \tau_{jl} \quad (8)$$

Since capital is fixed within the industry in the short run, marginal cost is increasing in output, and β_{ij} governs the slope of the industry supply curve. Industries that are more intensive in the fixed factor will have less elastic supply, and prices in these industries will be more responsive to demand shocks.

2.3 Aggregation, Factor Payments, and Income

To aggregate production to industry-level gross output, multiply both sides of (8) by q_{ijl} , and sum across destinations to get:

$$V_{ij} \equiv \sum_l p_{ijl} q_{ijl} = \frac{w_j^{\gamma_{ij}} P_j^{(1-\gamma_{ij})}}{1 - \beta_{ij}} \frac{Q_{ij}^{\frac{\beta_{ij}}{1-\beta_{ij}}}}{K_{ij}^{\frac{\beta_{ij}}{1-\beta_{ij}}}} \sum_l \tau_{jl} q_{ijl} = \frac{C_{ij}(Q_{ij})}{1 - \beta_{ij}} \quad (9)$$

One minor note in this aggregation is that, due to the iceberg assumption, total production Q_{ij} is the sum of iceberg-adjusted consumer demands, $\tau_{jl} q_{ijl}$.

Variable profits, which are distributed to owners of capital, who earn r_{ij} per unit of capital, are written as:

$$r_{ij} K_{ij} = \beta_{ij} V_{ij} \quad (10)$$

Similarly, payments to labor, and spending on intermediate inputs, are written as:

$$w_j L_{ij} = \gamma_{ij} (1 - \beta_{ij}) V_{ij} \quad (11)$$

$$P_j M_{ij} = (1 - \gamma_{ij}) (1 - \beta_{ij}) V_{ij} \quad (12)$$

Consumer income is the sum of factor payments, $Y_j^c = w_j L_j + \sum_i r_{ij} K_{ij}$. Total spending in the economy, Y_j , is the sum of factor payments and input payments:

$$Y_j = w_j L_j + \sum_i r_{ij} K_{ij} + P_j \sum_i M_{ij} \quad (13)$$

Since inputs are purchased in the same proportions as final goods, spending on industry i in country j is $\theta_{ij}Y_j$, which is split across origins according to λ_{ikj} . With balanced trade, $Y_j = \sum_i V_{ij}$. In the long run, capital is perfectly mobile across industries within a country, and the rental rate is equalized across industries, $r_{ij} = \tilde{r}_j$ for all i , with the total stock of capital $K_j = \sum_i K_{ij}$ fixed. We now manipulate these equations to solve for changes in prices and welfare.

3 Exact Changes in Prices

In the first task of the analysis, we derive the change in industry-level and aggregate prices. To facilitate analysis for both the short and long run, we write

$$p_{ijl} = \tau_{jl} w_j^{\gamma_{ij}} P_j^{(1-\gamma_{ij})} a_{ij} \quad (14)$$

where, in the short-run model (capital fixed), a_{ij} is:

$$a_{ij} = \frac{Q_{ij}^{\frac{\beta_{ij}}{1-\beta_{ij}}}}{(1-\beta_{ij}) K_{ij}^{\frac{\beta_{ij}}{1-\beta_{ij}}}} = \frac{1}{(1-\beta_{ij})^{1-\beta_{ij}}} \left(\frac{V_{ij}}{w_j^{\gamma_{ij}} P_j^{1-\gamma_{ij}} K_{ij}} \right)^{\beta_{ij}} \quad (15)$$

In the long-run model, when all factors can adjust, a_{ij} is written as:

$$a_{ij} = \left(\frac{\tilde{r}_j}{w_j^{\gamma_{ij}} P_j^{1-\gamma_{ij}}} \right)^{\beta_{ij}} \quad (16)$$

Without loss of generality, we assume that labor in j is the numeraire, $w_j = 1$, and that we aim to assess the welfare effects of an arbitrary trade shock for country j . Throughout, we denote the value of a variable after the shock with a prime, and use a tilde to denote outcomes in the long-run model.

3.1 Short Run

Following the steps in ACR, in Appendix A we show that the exact change in the industry i price index in j is

$$\frac{P'_{ij}}{P_{ij}} = \left(\frac{\lambda'_{ijj}}{\lambda_{ijj}} \right)^{\frac{1}{\sigma_{ij}-1}} \left(\frac{P'_j}{P_j} \right)^{1-\gamma_{ij}} \frac{a'_{ij}}{a_{ij}} \quad (17)$$

In the short run, the exact change in a_{ij} follows from (15), which, after aggregating across industries using the Cobb-Douglas aggregator, leads to the following simple characterization of changes in aggregate prices:

$$\frac{P'_j}{P_j} = \left(\prod_i^I \left(\frac{\lambda'_{ijj}}{\lambda_{ijj}} \right)^{\frac{\theta_{ij}}{\sigma_{ij}-1}} \left(\frac{V'_{ij}}{V_{ij}} \right)^{\theta_{ij}\beta_{ij}} \right)^{\frac{1}{1-\sum_i^I \theta_{ij}(1-\beta_{ij})(1-\gamma_{ij})}} \quad (18)$$

From (10), and since capital is fixed within each industry, the exact change in the nominal return to capital in industry i is simply the exact change in the value of output in that industry:

$$\frac{r'_{ij}}{r_{ij}} = \frac{V'_{ij}}{V_{ij}} \quad (19)$$

Substituting into the exact change in aggregate prices, we get:

$$\frac{P'_j}{P_j} = \left(\prod_i^I \left(\frac{\lambda'_{ijj}}{\lambda_{ijj}} \right)^{\frac{\theta_{ij}}{\sigma_{ij}-1}} \left(\frac{r'_{ij}}{r_{ij}} \right)^{\theta_{ij}\beta_{ij}} \right)^{\frac{1}{1-\sum_i^I \theta_{ij}(1-\beta_{ij})(1-\gamma_{ij})}} \quad (20)$$

Thus, exact changes in prices under specific factors are determined by the standard formula, and whether capital in each industry is getting more or less expensive relative to the numeraire (labor) in j .

3.2 Long Run

In the long run, it is straightforward to use (16) to derive a similar form (this is also summarized in the appendix of ACR):

$$\frac{\tilde{P}'_j}{\tilde{P}_j} = \left(\prod_i^I \left(\frac{\lambda'_{ijj}}{\lambda_{ijj}} \right)^{\frac{\theta_{ij}}{\sigma_{ij}-1}} \left(\frac{\tilde{r}'_j}{\tilde{r}_j} \right)^{\theta_{ij}\beta_{ij}} \right)^{\frac{1}{1-\sum_i^I \theta_{ij}(1-\beta_{ij})(1-\gamma_{ij})}} \quad (21)$$

Thus, the returns to capital, and whether they are industry-specific, guide the adjustment in prices, and therefore welfare.

To solve for the change in the long-run rental rate, we use a technique articulated in the appendix of ACR. Since both the total stock of capital and labor are fixed, the aggregate capital-labor ratio is the sum of the industry-level ratios, weighted by the industry i share

of labor employed within j , δ_{ij} . Using the first-order conditions in (10) and (11), we have:

$$\frac{K_j}{L_j} = \sum_i \delta_{ij} \frac{K_{ij}}{L_{ij}} = \frac{w_j}{\tilde{r}_j} \sum_i \delta_{ij} \frac{\beta_{ij}}{(1 - \beta_{ij})\gamma_{ij}} \quad (22)$$

where

$$\delta_{ij} = \frac{L_{ij}}{L_j} = \frac{(1 - \beta_{ij})\gamma_{ij}V_{ij}}{\sum_h (1 - \beta_{hj})\gamma_{hj}V_{hj}} \quad (23)$$

Combining these equations, we get:

$$\frac{K_j}{L_j} \frac{\tilde{r}_j}{w_j} = \frac{\sum_i \beta_{ij}V_{ij}}{\sum_h (1 - \beta_{hj})\gamma_{hj}V_{hj}} \quad (24)$$

which is simply the ratio of total capital payments to total labor payments. Since K_j/L_j is fixed and $w_j = 1$, exact changes in long-run rental rates are:

$$\frac{\tilde{r}'_j}{\tilde{r}_j} = \frac{\frac{\sum_i \beta_{ij}V'_{ij}}{\sum_h (1 - \beta_{hj})\gamma_{hj}V'_{hj}}}{\frac{\sum_i \beta_{ij}V_{ij}}{\sum_h (1 - \beta_{hj})\gamma_{hj}V_{hj}}} \quad (25)$$

This yields the long-run change in aggregate prices:

$$\frac{\tilde{P}'_j}{P_j} = \left(\prod_i \left(\frac{\lambda'_{ijj}}{\lambda_{ijj}} \right)^{\frac{\theta_{ij}}{\sigma_{ij}-1}} \left(\frac{\frac{\sum_h \beta_{hj}V'_{hj}}{\sum_h (1 - \beta_{hj})\gamma_{hj}V'_{hj}}}{\frac{\sum_h \beta_{hj}V_{hj}}{\sum_h (1 - \beta_{hj})\gamma_{hj}V_{hj}}} \right)^{\theta_{ij}\beta_{ij}} \right)^{\frac{1}{1 - \sum_i \theta_{ij}(1 - \beta_{ij})(1 - \gamma_{ij})}} \quad (26)$$

Overall, we can compare changes in aggregate prices in the short and long run in (20) and (26) by observing that the difference is entirely driven by whether changes in gross output are pooled across all industries (in 26), or whether they specifically affect the returns to capital within an industry (in 20). While both measures are readily calculated, since they don't necessarily follow trade shares, we need to add national accounts data to characterize the gains from trade. We now use these equations to characterize the changes in welfare for factor owners when the economy moves to autarky.

4 Return to Autarky

Suppose that the economy is in a long-run equilibrium, and a trade shock brings country j into autarky. What are the impacts on welfare across factor owners? And how do we

characterize the transition between fixed capital and perfectly mobile capital? The key to the analysis is invoking, many times, that the value of output must equal expenditure within each industry in autarky. This pins down exact changes in income and rental rates as a function of observables at baseline and model primitives. Throughout, we use the superscript a to denote outcomes in autarky.

4.1 Changes in Income

Since labor in j is the numeraire, and the total supply of labor is fixed, exact changes in total spending are inversely related to exact changes in the share of labor payments in total spending, s_j .¹

$$\frac{Y'_j}{Y_j} = \frac{s_j}{s'_j} = \frac{\frac{\sum_i (1-\beta_{ij}) \gamma_{ij} V_{ij}}{\sum_i V_{ij}}}{\frac{\sum_i (1-\beta_{ij}) \gamma_{ij} V'_{ij}}{\sum_i V'_{ij}}} \quad (27)$$

In autarky, output must equal expenditure within each industry, $V_{ij}^a = \theta_{ij} Y_j^a$. Thus, the change in total spending when moving to autarky is:

$$\frac{Y_j^a}{Y_j} = \frac{s_j}{s_j^a} = \frac{\frac{\sum_i (1-\beta_{ij}) \gamma_{ij} V_{ij}}{\sum_i V_{ij}}}{\sum_i (1-\beta_{ij}) \gamma_{ij} \theta_{ij}} \quad (28)$$

Changes in nominal income are the same in both the specific factors and long-run models, since expenditure shares on goods and factor payments are tightly linked in autarky. This is the first sense in which autarky is a useful benchmark: all differences between the short and long run will operate through prices.

4.2 Rental Rates in the Long Run

Also following from market clearing in autarky, the long-run change in the rental rate can be written as:²

$$\frac{\tilde{r}_j^a}{r_j} = \frac{\frac{\sum_h \beta_{hj} \theta_{hj}}{\sum_h (1-\beta_{hj}) \gamma_{hj} \theta_{hj}}}{\frac{\sum_h \beta_{hj} V_{hj}}{\sum_h (1-\beta_{hj}) \gamma_{hj} V_{hj}}} = \frac{B_j^a / (1 - B_j^a)}{B_j / (1 - B_j)} \quad (29)$$

¹This is also articulated in the appendix of ACR for long-run changes in total income, but it holds for short-run changes as well.

²This follows from substituting $V_{ij}^a = \theta_{ij} Y_j^a$ into (25), which yields $\frac{\tilde{r}_j^a}{r_j} = \frac{\sum_h (1-\beta_{hj}) \gamma_{hj} V_{hj}}{\sum_h (1-\beta_{hj}) \gamma_{hj} \theta_{hj}} \cdot \frac{\sum_h \beta_{hj} \theta_{hj}}{\sum_h \beta_{hj} V_{hj}}$, and rearranging.

where

$$B_j^a \equiv \frac{r_j^a K_j}{r_j^a K_j + w_j^a L_j} = \frac{\sum_h \beta_{hj} \theta_{hj}}{\sum_h \beta_{hj} \theta_{hj} + \sum_h \theta_{hj} (1 - \beta_{hj}) \gamma_{hj}} \quad (30)$$

$$B_j \equiv \frac{r_j K_j}{r_j K_j + w_j L_j} = \frac{\sum_h \beta_{hj} V_{hj}}{\sum_h \beta_{hj} V_{hj} + \sum_h (1 - \beta_{hj}) \gamma_{hj} V_{hj}} \quad (31)$$

B_j and B_j^a are the shares of capital payments in value-added, at baseline and in autarky, respectively. These factor income shares are central to weighting the factor-related components of the gains from trade formulas, a result that we articulate shortly. It is important to note that both shares can be calculated using output and factor intensities. For autarky, they are calculated by again invoking that the value of output and expenditures must be equal within all industries.

Equation (29) has a familiar Stolper-Samuelson interpretation. The long-run rental rate rises relative to the wage when moving to autarky if and only if $B_j^a > B_j$, that is, if consumption is tilted toward capital-intensive industries relative to production at baseline. Put differently, capital owners gain relative to workers from autarky in the long run in countries whose trade embodies net exports of labor services, and lose relative to workers in countries whose trade embodies net exports of capital services.

4.3 Rental Rates with Specific Factors

With specific factors, the exact change in the rental rate within each industry is the exact change in industry output, from (19). By rearranging, and using the exact change in total spending in (28), we get:³

$$\frac{r_{ij}^a}{r_j} = \frac{\beta_{ij} \theta_{ij}}{\beta_{ij} V_{ij}} \cdot \frac{\sum_h (1 - \beta_{hj}) \gamma_{hj} V_{hj}}{\sum_h (1 - \beta_{hj}) \gamma_{hj} \theta_{hj}} \quad (32)$$

Multiplying and dividing by the ratio of total capital payments in autarky and at baseline links this formula tightly to the long-run change in rental rates:

$$\frac{r_{ij}^a}{r_j} = \underbrace{\frac{\beta_{ij} \theta_{ij}}{\sum_h \beta_{hj} \theta_{hj}}}_{\kappa_{ij}^a / \kappa_{ij}} \cdot \underbrace{\frac{\sum_h (1 - \beta_{hj}) \gamma_{hj} V_{hj}}{\sum_h (1 - \beta_{hj}) \gamma_{hj} \theta_{hj}}}_{\tilde{r}_j^a / r_j} \cdot \frac{\sum_h \beta_{hj} \theta_{hj}}{\sum_h \beta_{hj} V_{hj}} = \frac{\kappa_{ij}^a}{\kappa_{ij}} \cdot \frac{\tilde{r}_j^a}{r_j} \quad (33)$$

³This is derived using $\frac{r_{ij}^a}{r_j} = \frac{V_{ij}^a}{V_{ij}} = \frac{\theta_{ij} Y_j^a}{V_{ij}} = \frac{\theta_{ij}}{V_{ij}} \cdot Y_j \frac{Y_j^a}{Y_j}$ and $Y_j = \sum_h V_{hj}$.

where the shares of industry i in total capital payments in country j , in autarky and at baseline, respectively, are written as:

$$\kappa_{ij}^a = \frac{\beta_{ij}\theta_{ij}}{\sum_h \beta_{hj}\theta_{hj}}, \quad \kappa_{ij} = \frac{\beta_{ij}V_{ij}}{\sum_h \beta_{hj}V_{hj}} \quad (34)$$

Both κ_{ij} and κ_{ij}^a can be calculated using factor shares and initial consumption and production data. Equation (33) shows that the short-run return to capital in industry i is the long-run return, scaled up or down by the exact change in that industry's share of total capital payments. Since total capital payments in autarky are the same whether or not capital is mobile, the short-run rental rates simply redistribute the long-run capital income across industries. Industries whose output share is high relative to their consumption share at baseline, which are typically net exporters, see $\kappa_{ij}^a < \kappa_{ij}$, and their capital owners are hurt relative to the long run. Industries that are net importers at baseline see the opposite.

4.4 Changes in Prices: Three Components

To finalize the gains from trade formula, define $Z_j = 1 - \sum_i \theta_{ij} (1 - \beta_{ij}) (1 - \gamma_{ij})$ as the share of value-added in gross output for country j in autarky. Substituting (33) into (20), the exact change in prices under specific factors is the long-run change in prices adjusted by a misallocation term:⁴

$$\frac{P_j^a}{P_j} = \underbrace{\prod_i \left(\frac{1}{\lambda_{ijj}} \right)^{\frac{\theta_{ij}}{(\sigma_{ij}-1)Z_j}}}_{\text{ACR, one factor w/ inputs}} \cdot \underbrace{\left(\frac{B_j^a/(1-B_j^a)}{B_j/(1-B_j)} \right)^{B_j^a}}_{\text{Multiple Factors}} \cdot \underbrace{\left(\prod_i \left(\frac{\kappa_{ij}^a}{\kappa_{ij}} \right)^{\kappa_{ij}^a} \right)^{B_j^a}}_{\text{SF Misallocation}} \quad (35)$$

In the long run, the misallocation term vanishes and we have the following formula:

$$\frac{\tilde{P}_j^a}{P_j} = \underbrace{\prod_i \left(\frac{1}{\lambda_{ijj}} \right)^{\frac{\theta_{ij}}{(\sigma_{ij}-1)Z_j}}}_{\text{ACR, one factor w/ inputs}} \cdot \underbrace{\left(\frac{B_j^a/(1-B_j^a)}{B_j/(1-B_j)} \right)^{B_j^a}}_{\text{Multiple Factors}} \quad (36)$$

⁴The exponents are simplified by noting that the exponent on the exact change in the rental rate within each industry can be linked to the share of capital payments in industry i in country j , and the share of capital payments in total value-added in country j , both in autarky:

$$\frac{\beta_{ij}\theta_{ij}}{1 - \sum_h \theta_{hj} (1 - \beta_{hj}) (1 - \gamma_{hj})} = \frac{\beta_{ij}\theta_{ij}}{\sum_h \beta_{hj}\theta_{hj}} \cdot \frac{\sum_h \beta_{hj}\theta_{hj}}{Z_j} = \kappa_{ij}^a \cdot B_j^a$$

where the second equality uses $Z_j = \sum_h \theta_{hj}\beta_{hj} + \sum_h \theta_{hj}(1 - \beta_{hj})\gamma_{hj}$.

For the remainder of the paper, we refer to the three components as ACR_j , HO_j , and MIS_j , respectively, so that $\tilde{P}_j^a/P_j = ACR_j \cdot HO_j$ and $P_j^a/P_j = ACR_j \cdot HO_j \cdot MIS_j$.

There are a few immediate points worth making regarding the price formulae, and the differences between them. First, looking at the long-run result in (36), while this result is mostly articulated in the appendix of ACR, the presentation here is more concise in that it reduces the multi-factor adjustment to a single statistic: the share of capital in total value-added. Further, it shows that this share, in autarky, is the key elasticity of prices, and therefore welfare, with respect to changes in rental rates in the long run. The HO_j term is greater than one if and only if $B_j^a > B_j$, and in these countries, the price index rises by more than the standard ACR formula would suggest, since capital becomes more expensive relative to labor.

Moving to the specific factors formula, B_j^a continues to weight the role of changes in factor prices, though now the change in rental rates includes both the long-run component (discussed in the prior paragraph), and the short-run misallocation, which is represented by the geometric average of exact changes in shares of capital payments, weighted by the share of capital payments of each industry in autarky. That is, misallocation is a function of: $\kappa_{ij}^a/\kappa_{ij}$, the exact change in industry i 's share of capital payments; geometrically weighted by κ_{ij}^a , the industry share of capital payments in autarky; all weighted by B_j^a , the share of capital payments in value-added in autarky.

Interestingly, the misallocation term *does not* depend on two variables that the standard gains from trade formula does. Indeed, the misallocation term has no direct relationship to the depth of trade at baseline (the domestic expenditure share), or to the elasticity of substitution. Indirectly, trade does matter, since misallocation is larger when there is a bigger difference between consumption shares and production shares of each industry at baseline. However, these consumption and production shares can be calculated without trade data. Instead, they can be calculated using industry-level output and consumption, and factor shares, all of which are available in national accounts.

The misallocation term also has a sharp characterization, which we summarize in the following proposition.

Proposition 1. *The misallocation term can be written as*

$$MIS_j = \exp \left(B_j^a \cdot D_{KL} \left(\kappa_j^a \parallel \kappa_j \right) \right), \quad D_{KL} \left(\kappa_j^a \parallel \kappa_j \right) \equiv \sum_i \kappa_{ij}^a \log \left(\frac{\kappa_{ij}^a}{\kappa_{ij}} \right) \quad (37)$$

where $D_{KL} \left(\kappa_j^a \parallel \kappa_j \right)$ is the Kullback-Leibler divergence of the distribution of capital payments across industries in autarky from the distribution at baseline. Accordingly, $MIS_j \geq 1$, with

equality if and only if $\kappa_{ij}^a = \kappa_{ij}$ for all i . Thus, for any country starting from a long-run equilibrium, the price increase from a return to autarky is always weakly larger when capital is specific than when capital is mobile.

The proof follows directly from taking logs of the misallocation term in (35), and from Gibbs' inequality, since both κ_j^a and κ_j are distributions that sum to one across industries. Proposition 1 has two immediate implications for welfare. Since labor is the numeraire, and nominal income in autarky is the same in the short and long run, both the real wage and aggregate real income are always weakly lower in the short run than in the long run when moving to autarky.

4.5 Welfare by Factor Owner

With these price formulae in hand, it is straightforward to characterize changes in real income by factor owner. Since labor is the numeraire, the change in the real wage is the inverse of the change in prices, P_j/P_j^a in the short run and $P_j/\tilde{P}_j^a$ in the long run. The change in the real income of owners of capital in industry i is:

$$\frac{W_{cap,ij}^a}{W_{cap,ij}} = \frac{r_{ij}^a/r_j}{P_j^a/P_j}, \quad \frac{\widetilde{W}_{cap,j}^a}{W_{cap,j}} = \frac{\tilde{r}_j^a/r_j}{\tilde{P}_j^a/P_j} \quad (38)$$

in the short and long run, respectively. Combining (35), (36), and (33), the short-run real return to capital in industry i relative to its long-run counterpart is simply $(\kappa_{ij}^a/\kappa_{ij})/MIS_j$. Thus, owners of capital in industry i prefer that capital be stuck in their industry when moving to autarky if and only if their share of total capital payments rises by more than the aggregate misallocation term. Finally, since nominal factor income in country j is $w_j L_j/(1 - B_j)$, and labor income is fixed by the choice of numeraire, the change in aggregate real factor income is:

$$\frac{W_{all,j}^a}{W_{all,j}} = \frac{(1 - B_j)/(1 - B_j^a)}{P_j^a/P_j}, \quad \frac{\widetilde{W}_{all,j}^a}{W_{all,j}} = \frac{(1 - B_j)/(1 - B_j^a)}{\tilde{P}_j^a/P_j} \quad (39)$$

We now turn to quantifying these welfare measures using readily available input-output data.

5 Data and Calibration

To quantify the formulas above, we require four sets of objects for each country: domestic expenditure shares by industry, λ_{ijj} ; expenditure shares across industries, θ_{ij} ; factor and input intensities, β_{ij} and γ_{ij} ; and elasticities of substitution, σ_{ij} . Recall from Section 4 that the elasticities and domestic expenditure shares are needed only for the ACR component. The factor proportions and misallocation terms require only industry-level output, consumption, and factor shares.

Our primary data source is the World Input-Output Database (WIOD, [Timmer et al., 2015](#)). For our analysis of 2014, we use the 2016 release of WIOD, which covers 43 countries and a rest-of-the-world aggregate from 2000 to 2014. For our analysis over time, we use the Long-Run WIOD ([Woltjer et al., 2021](#)), which covers 1965 to 2000 for a smaller set of countries and a coarser set of industries. To ensure consistency across the two periods, we aggregate the 2016 release to the 23 industries used in the Long-Run WIOD.

We calibrate the model as follows. Domestic expenditure shares, λ_{ijj} , are calculated as the share of total domestic use (final and intermediate) of industry i goods in country j that is sourced from domestic producers. Industry expenditure shares, θ_{ij} , are calculated as gross consumption shares by industry within each country, that is, the share of industry i in total final and intermediate use in country j . This is consistent with the model, in which inputs are purchased in the same proportions as final goods. Factor and input intensities, β_{ij} and γ_{ij} , are calculated using the Socio-Economic Accounts that accompany WIOD. Specifically, $(1 - \beta_{ij})\gamma_{ij}$ is the share of labor compensation in gross output, and $(1 - \beta_{ij})(1 - \gamma_{ij})$ is the share of intermediate inputs in gross output. To limit the influence of year-to-year noise, we set β_{ij} and γ_{ij} to their median values over 2000–2014 for each country and industry, and set $\beta_{ij} = 0.01$ in the few cases where measured capital payments are negative. Capital payments are then the residual of the gross output identity, so that capital is interpreted as anything in gross output that is not labor payments or purchased inputs. This includes returns to physical capital, but also returns to land, natural resources, and other fixed factors, as well as pure profits. Finally, in our baseline results, we set the elasticity of substitution to $\sigma_{ij} = 4$ for all industries and countries. Recall that this elasticity affects only the ACR component, and not the factor proportions or misallocation terms.

Two limitations of our current calibration are worth noting. First, we have not yet dealt with trade deficits. Our formulas assume balanced trade at baseline, and in future drafts, we plan to follow the literature in first purging deficits from the data before computing counterfactuals. Second, for the analysis from 1965 to 2014, we use the same invariant factor intensities, the 2000–2014 medians described above, for the entire exercise. For country-

industry pairs in the Long-Run WIOD without a match in the 2016 release, we use the cross-country average for that industry. Thus, all variation over time in our results comes from changes in trade shares, output, and consumption shares, rather than from changes in factor intensity. We are currently merging supplementary data for longer coverage of capital, labor, and input shares, at least for the US.

6 A Return to Autarky in 2014

We begin by calculating the effects of a return to autarky starting from 2014, the most recent year in our data. We first present aggregate results for all 43 countries, and then turn to industry-level results for capital owners.

6.1 Aggregate Results

Table 1 presents the decomposition of the change in prices from (35) and (36), along with changes in the long-run rental rate and real income by factor owner. Several patterns stand out. First, the long-run cost of autarky is substantial in most countries, and dominated by the ACR component. On average, long-run prices rise by 34% when moving to autarky (25% at the median), with large increases for small, open economies such as Malta (137%), Luxembourg (105%), and Hungary (86%), and smaller increases for large economies such as the US (5.8%), Brazil (3.5%), and China (6.6%).

Second, the factor proportions term, HO_j , is close to one in most countries, averaging 0.996. However, it is informative about the factor content of trade. The HO_j term is noticeably below one in Norway (0.846), Russia (0.935), Indonesia (0.965), Australia (0.974), and Luxembourg (0.974). In these countries, autarky reduces the share of capital in value-added, and the long-run rental rate falls relative to the wage. Given that capital in our calibration includes returns to natural resources, it is not surprising that many of these countries are large exporters of resource-intensive goods. In Norway, the fall in the rental rate is large enough that long-run prices fall relative to the wage when moving to autarky, and workers gain 1.9% in real terms in the long run, while capital owners lose 35%. At the other end, the HO_j term is above one in Malta (1.085), Japan (1.048), India (1.034), and Mexico (1.032), where autarky shifts demand toward capital-intensive industries.

Third, the misallocation term, MIS_j , is fairly modest in most countries. Prices rise an additional 3.7% on average in the short run relative to the long run (2.4% at the median). In the US, the additional increase is only 0.6%, on top of a 5.8% long-run increase. One exception is China, where prices rise 6.6% in the long run, and an additional 2.6% in the short

run. Thus, for China, misallocation through specific factors raises the (log) price increase from autarky by roughly 40%, while for the US, the corresponding figure is roughly 10%. The largest misallocation terms are in Lithuania (14.6%), Ireland (12.8%), Korea (11.0%), and Norway (10.2%), all of which have production structures that are quite different from their consumption structures. Across all countries, the misallocation term accounts for 16% of the log change in short-run prices on average, and 10% at the median.⁵

Finally, the last four columns of Table 1 show changes in real income. In the long run, aggregate real factor income falls by 23% on average when moving to autarky, and by 26% in the short run. In the US, the corresponding figures are 4.9% and 5.5%, respectively. Long-run losses for capital owners and workers are broadly similar in most countries, which is consistent with the HO_j term being close to one. However, there are notable differences in a few countries. In China, capital owners lose 5.3% in the long run, while workers lose 6.2%. In Japan, capital owners lose 3.5%, while workers lose 13.2%. In all cases, the long-run change in aggregate real factor income is a weighted average of the changes for workers and capital owners, with weights $1 - B_j$ and B_j .

6.2 Industry-Level Results

While the aggregate costs of specific factors are modest, they may mask much larger effects for capital owners in individual industries. Table 2 presents industry-level results for the US and China, sorted by the change in real income for US capital owners in the short run. For each country, we report the change in the rental rate under specific factors, r_{ij}^a/r_j , and the change in real income for owners of capital in that industry, $W_{cap,ij}^a/W_{cap,ij}$. The last row reports the corresponding long-run values, which are common to all industries. Corresponding results for Canada and Mexico are reported in Appendix Table 3.

The dispersion in short-run returns is dramatic. In the long run, capital owners in the US lose 4.3% of real income when moving to autarky. In the short run, however, capital owners in US textiles, leather and footwear experience a 158% increase in real income. This reflects the fact that the US consumes far more textiles than it produces, so that the share of textiles in total capital payments nearly triples in autarky. Owners of capital in electricity, gas and water supply (17%), basic metals (11%), other non-metallic minerals (11%), and mining (9%) also gain in the short run. On the other side, owners of capital in machinery lose 46% of real income, and owners of capital in transport equipment lose 31%. These are industries in which the US has a large production share relative to its consumption share. In total, capital owners in 14 of the 23 US industries would gain in real terms from autarky

⁵The average is inflated by Norway, where long-run prices fall relative to the wage, so that the misallocation term exceeds the total log change in short-run prices. Excluding Norway, the average is 13%.

Table 1: Returns to Autarky by Country, 2014

Country	Price Index Decomposition					Rental Rate and Real Income				
	ACR_j	HO_j	MIS_j	Long Run		Long Run			Short Run	
				$\frac{\tilde{P}_j^a}{P_j}$	$\frac{P_j^a}{P_j}$	$\frac{\tilde{r}_j^a}{r_j}$	$\frac{\tilde{W}_{cap,j}^a}{W_{cap,j}}$	$\frac{\tilde{W}_{lab,j}^a}{W_{lab,j}}$	$\frac{\tilde{W}_{all,j}^a}{W_{all,j}}$	$\frac{W_{all,j}^a}{W_{all,j}}$
AUS	1.100	0.974	1.028	1.071	1.101	0.937	0.874	0.933	0.909	0.884
AUT	1.305	1.003	1.017	1.309	1.331	1.008	0.770	0.764	0.766	0.754
BEL	1.535	1.002	1.021	1.538	1.570	1.005	0.654	0.650	0.651	0.638
BGR	1.455	1.000	1.042	1.455	1.516	1.000	0.687	0.687	0.687	0.660
BRA	1.055	0.981	1.024	1.035	1.060	0.961	0.929	0.966	0.948	0.926
CAN	1.191	0.980	1.023	1.167	1.194	0.951	0.815	0.857	0.840	0.821
CHE	1.191	0.991	1.009	1.180	1.191	0.975	0.826	0.848	0.840	0.832
CHN	1.061	1.005	1.026	1.066	1.094	1.010	0.947	0.938	0.943	0.919
CYP	1.330	0.980	1.067	1.303	1.390	0.954	0.732	0.768	0.752	0.705
CZE	1.538	1.005	1.018	1.545	1.574	1.010	0.654	0.647	0.650	0.639
DEU	1.190	1.012	1.008	1.204	1.214	1.032	0.857	0.830	0.840	0.834
DNK	1.314	1.005	1.008	1.321	1.332	1.015	0.768	0.757	0.761	0.755
ESP	1.197	1.002	1.018	1.199	1.221	1.005	0.838	0.834	0.835	0.820
EST	1.705	1.005	1.034	1.713	1.772	1.011	0.590	0.584	0.587	0.567
FIN	1.204	1.004	1.015	1.209	1.228	1.011	0.836	0.827	0.830	0.818
FRA	1.424	1.002	1.011	1.427	1.443	1.006	0.705	0.701	0.702	0.695
GBR	1.132	1.001	1.004	1.133	1.137	1.002	0.885	0.883	0.883	0.880
GRC	1.179	0.987	1.034	1.164	1.203	0.973	0.836	0.859	0.848	0.820
HRV	1.291	1.013	1.019	1.307	1.331	1.039	0.795	0.765	0.775	0.761
HUN	1.871	0.992	1.043	1.856	1.935	0.982	0.529	0.539	0.534	0.512
IDN	1.111	0.965	1.051	1.072	1.127	0.932	0.869	0.933	0.900	0.856
IND	1.087	1.034	1.040	1.124	1.169	1.067	0.949	0.890	0.920	0.884
IRL	1.688	0.991	1.128	1.673	1.887	0.982	0.587	0.598	0.593	0.525
ITA	1.121	1.015	1.014	1.137	1.154	1.034	0.909	0.879	0.892	0.880
JPN	1.100	1.048	1.020	1.153	1.176	1.112	0.965	0.868	0.908	0.890
KOR	1.264	1.023	1.110	1.293	1.435	1.060	0.820	0.774	0.791	0.712
LTU	1.523	1.011	1.146	1.539	1.763	1.021	0.663	0.650	0.656	0.573
LUX	2.102	0.974	1.046	2.047	2.142	0.941	0.460	0.489	0.475	0.454
LVA	1.473	1.002	1.077	1.476	1.590	1.005	0.681	0.678	0.679	0.630
MEX	1.434	1.032	1.037	1.479	1.535	1.049	0.709	0.676	0.697	0.672
MLT	2.182	1.085	1.063	2.368	2.517	1.189	0.502	0.422	0.457	0.430
NLD	1.430	0.976	1.026	1.395	1.432	0.934	0.669	0.717	0.699	0.681
NOR	1.160	0.846	1.102	0.981	1.081	0.635	0.647	1.019	0.841	0.763
POL	1.260	0.977	1.012	1.232	1.247	0.954	0.774	0.812	0.793	0.784
PRT	1.238	1.006	1.024	1.245	1.275	1.015	0.815	0.803	0.808	0.789
ROU	1.227	0.978	1.023	1.201	1.228	0.959	0.799	0.833	0.815	0.797
RUS	1.127	0.935	1.047	1.053	1.103	0.837	0.795	0.950	0.884	0.844
SVK	1.787	1.006	1.069	1.798	1.923	1.012	0.563	0.556	0.560	0.523
SVN	1.397	0.988	1.012	1.380	1.396	0.966	0.700	0.724	0.716	0.707
SWE	1.205	1.001	1.017	1.205	1.226	1.002	0.831	0.830	0.830	0.816
TUR	1.166	0.981	1.036	1.143	1.184	0.969	0.848	0.875	0.858	0.828
TWN	1.333	0.985	1.014	1.313	1.331	0.898	0.684	0.761	0.750	0.739
USA	1.052	1.006	1.006	1.058	1.064	1.013	0.957	0.945	0.951	0.945
Mean	1.343	0.996	1.037	1.339	1.391	0.988	0.761	0.775	0.769	0.743
Median	1.260	1.002	1.024	1.245	1.275	1.005	0.795	0.803	0.793	0.763

Notes: Return to autarky from 2014, 2016 release of WIOD aggregated to 23 industries. ACR_j , HO_j , and MIS_j are the components of (35), so that $\tilde{P}_j^a/P_j = ACR_j \cdot HO_j$ and $P_j^a/P_j = ACR_j \cdot HO_j \cdot MIS_j$. $\tilde{r}_j^a/r_j$ is the long-run change in the rental rate relative to the wage. $\tilde{W}_{cap}$, $\tilde{W}_{lab}$, and $\tilde{W}_{all}$ are long-run changes in real income for capital owners, workers, and in aggregate; W_{all} is the short-run change in aggregate real income. Aggregate real income is real factor income, $\left[(1 - B_j)/(1 - B_j^a) \right] / (P_j^a/P_j)$.

in the short run, even though capital owners as a whole lose in the long run.

The results for China tell a similar story, though with a different set of winners and losers. Owners of capital in Chinese mining gain 29% in the short run when moving to autarky, reflecting China’s large imports of raw materials, while capital owners in chemicals, hotels and restaurants, and several service industries gain between 5% and 8%. The largest losers are in machinery (−39%), transport equipment (−33%), post and telecommunications (−20%), electrical and optical equipment (−19%), and textiles (−18%). Notably, textiles is at the top of the distribution in the US, and near the bottom in China.

In both countries, construction is by far the largest loser in the short run, with capital owners losing 79% of real income in the US and 91% in China. We caution against over-interpreting this result. Construction, and to a lesser extent machinery and transport equipment, are industries whose output is largely used for investment. Since the model has no investment, the gross consumption shares θ_{ij} for these industries may understate their long-run demand, which in turn exaggerates the fall in their share of capital payments in autarky.

7 Changes in the Returns to Autarky, 1965–2014

A key advantage of our sufficient statistics is that they can be computed with readily available data over long periods of time. In this section, we use the Long-Run WIOD, spliced with the 2016 release, to calculate the returns to autarky in each year from 1965 to 2014. We ask two questions. First, has misallocation from specific capital become more or less important over time, relative to the other components of the gains from trade? Second, how have the returns to industry-specific capital changed over time, and do any of these changes align with known policies or shocks? As noted in Section 5, the results in this section hold factor intensities fixed at their calibrated values, and so they should be interpreted as reflecting changes in the pattern of trade, production, and consumption, rather than changes in technology.

7.1 Components of the Gains from Trade over Time

Figure 1 plots each component of (35) and (36) for the US and China from 1965 to 2014. For the US, the short-run price increase from autarky nearly doubles over the period, from about 3.6% in 1965 to 6.4% in 2014, with a peak of nearly 8% in 2008. Most of this increase is driven by the ACR component, which rises from about 1.5% to 5.2%, as the US has become steadily more open. The misallocation term, in contrast, is fairly constant over the entire period, fluctuating between roughly 0.6% and 1.2%. As a consequence, the importance of

Table 2: Industry-Level Returns to Autarky for Capital Owners, US and China, 2014

Industry	United States		China	
	$\frac{r_{ij}^a}{r_j}$	$\frac{W_{cap,ij}^a}{W_{cap,ij}}$	$\frac{r_{ij}^a}{r_j}$	$\frac{W_{cap,ij}^a}{W_{cap,ij}}$
Construction	0.220	0.207	0.095	0.087
Machinery, Nec	0.574	0.540	0.666	0.609
Transport Equipment	0.734	0.690	0.736	0.673
Pulp, Paper, Printing and Publishing	0.825	0.775	1.156	1.056
Post and Telecommunications	0.860	0.809	0.878	0.802
Transport and Storage	0.974	0.915	1.095	1.001
Wholesale and Retail Trade	0.975	0.917	0.975	0.892
Real Estate, Renting and Business Act.	1.005	0.944	1.039	0.950
Manufacturing, Nec; Recycling	1.037	0.975	1.028	0.940
Coke, Refined Petroleum and Nuclear Fuel	1.066	1.002	1.151	1.052
Electrical and Optical Equipment	1.068	1.004	0.884	0.808
Financial Intermediation	1.068	1.004	1.152	1.053
Agriculture, Hunting, Forestry and Fishing	1.083	1.018	1.114	1.018
Community Social and Personal Services	1.100	1.034	1.161	1.062
Food, Beverages and Tobacco	1.100	1.034	1.126	1.030
Hotels and Restaurants	1.113	1.046	1.183	1.082
Chemicals and Chemical Products	1.119	1.052	1.165	1.065
Rubber and Plastics	1.137	1.069	1.060	0.969
Mining and Quarrying	1.161	1.091	1.408	1.287
Other Non-Metallic Mineral	1.175	1.105	1.105	1.010
Basic Metals and Fabricated Metal	1.176	1.105	1.107	1.012
Electricity, Gas and Water Supply	1.241	1.166	1.147	1.049
Textiles, Leather and Footwear	2.741	2.577	0.902	0.825
Long run (all industries): $\tilde{r}_j^a/r_j, \tilde{W}_{cap}^a/W_{cap}$	1.013	0.957	1.010	0.947

Notes: Short-run results for a return to autarky starting from 2014. r_{ij}^a/r_j is the change in the rental rate in industry i relative to the wage under specific factors, from (33), and $W_{cap,ij}^a/W_{cap,ij}$ is the corresponding change in real income for owners of capital in industry i . Industries are sorted by $W_{cap,ij}^a/W_{cap,ij}$ for United States. The last row reports the long-run change in the rental rate and real income for capital owners, which is common across industries.

misallocation has fallen as the gains from variety have risen. In 1965, the misallocation term was roughly two-thirds the size of the ACR component in logs, and accounted for more than a quarter of the short-run price increase. By 2014, it was roughly one-eighth the size of the ACR component. Interestingly, the factor proportions term for the US is not trivial, and is highest in the late 1970s and mid-2000s, both periods of high energy prices. This suggests that in these periods, a return to autarky would have shifted demand toward domestic industries that are capital-intensive in our calibration, such as mining, whose capital payments include returns to natural resources.

The results for China are quite different. Prior to 1980, when China was relatively closed, the misallocation term was roughly as large as the traditional ACR calculation, with both between about 1 and 3%. Over the following decades, the ACR component rose dramatically, peaking at over 10% in the mid-2000s before falling to about 6% by 2014, reflecting China's rapid integration into global markets following its accession to the WTO in 2001, and the subsequent decline in its trade openness after the global financial crisis. Throughout this period, the misallocation term also rose, from about 1.5% in the 1960s to roughly 2.5–3% by the late 2000s. Thus, unlike the US, misallocation through specific factors remains an important part of the cost of autarky for China. The factor proportions term is also informative about the evolution of China's comparative advantage. Prior to the mid-1980s, HO_j was below one, while it has been at or above one since the late 1980s, consistent with China's transition toward net exports of labor-intensive goods.

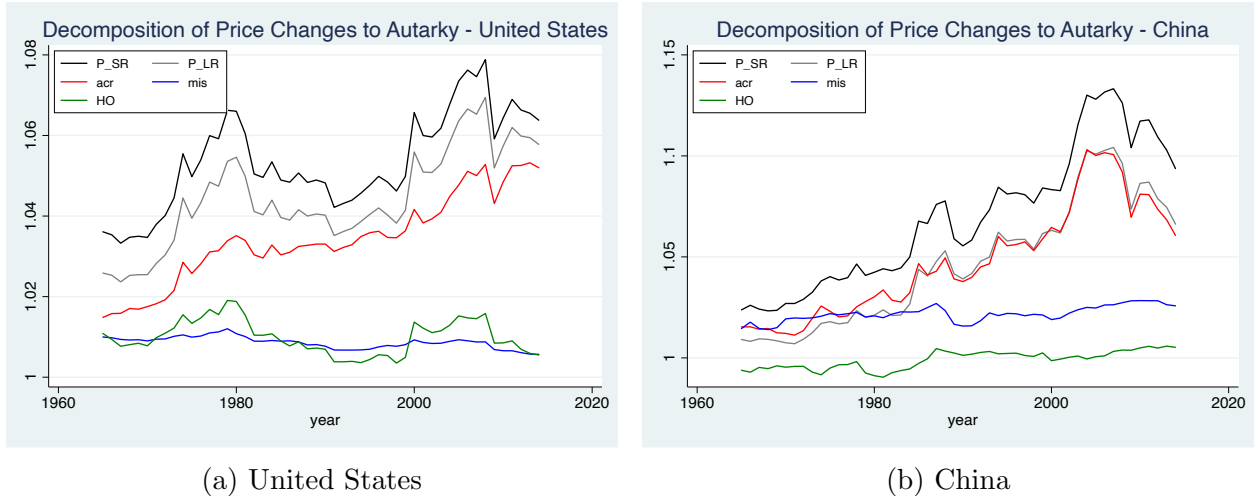


Figure 1: Decomposition of Price Changes from a Return to Autarky, 1965–2014

Notes: Each line plots a component of (35) and (36) for a return to autarky from each year. P_SR and P_LR are the short and long-run changes in the price index, P_j^a/P_j and $\tilde{P}_j^a/P_j$. acr , HO , and mis are the ACR, factor proportions, and misallocation components, respectively. Factor intensities are held fixed at their 2000–2014 medians.

7.2 Industry-Specific Returns over Time

We now turn to the returns for capital owners in individual industries. Figures 2 to 4 plot the short-run change in real income for capital owners from a return to autarky in each year, for selected industries in the US and China. In each panel, the thick red line is the highlighted industry, the gray lines are all other industries, and the black line is the long-run change in real income for capital owners, which is common across industries. Table 4 in the Appendix reports the US values for all industries in selected years.

Overall, we find dramatic changes in industry returns to autarky over the period, with some industries switching from “anti-autarky” to “pro-autarky” under specific factors, and vice versa. The most striking example is textiles, leather and footwear in Figure 2. In the US, the short-run return to autarky for textile capital owners was modestly positive, around 15%, through the mid-1970s, and rose gradually to about 40% by the early 1990s. Following the implementation of the Agreement on Textiles and Clothing in the mid-1990s, which set a path for the elimination of textile and apparel quotas under the Multi-Fiber Arrangement, the return to autarky rose steadily, and then accelerated sharply after the remaining quotas were removed in 2005, peaking at a gain of roughly 180% around 2008. The mirror image is visible in China, where textile capital owners have always been hurt by autarky in the short run, but where these losses were largest, at over 30%, around the time quotas were eliminated in 2005 (Khandelwal et al., 2013). Since then, the losses to Chinese textile capital owners from autarky have become smaller, consistent with a larger share of Chinese textile production being absorbed domestically.

Other industries show similarly large swings. In the US, capital owners in electrical and optical equipment moved from losing roughly 20–30% from autarky in the 1960s through the 1980s to roughly zero by 2014. In contrast, capital owners in pulp, paper, printing and publishing moved from roughly zero in the 1960s to losses of more than 20% in the 2000s, and capital owners in wholesale and retail trade, and transport and storage, moved from small gains to losses of 8–9%. Outside of textiles, the returns for US mining are the most volatile, with gains from autarky of 27% in 1975 and 36% in 2005, both periods of high oil prices and substantial energy imports, and a much smaller gain of 9% in 2014, which likely reflects the expansion of domestic oil and gas production after 2008. Returns to capital owners in basic metals, agriculture, and transport equipment were comparatively stable.

For China, the largest change is in mining, where the return to autarky for capital owners was close to zero until the late 1990s, and then rose sharply to a gain of more than 30% by the early 2010s, coincident with the rapid growth in China’s imports of raw materials. In electrical and optical equipment, capital owners moved from small losses in the 1960s and 1970s to losses of roughly 15–25% from the 1990s onward, as China became a major exporter

of electronics. Capital owners in basic metals experienced gains from autarky of 10–20% in the 1970s, which fell to roughly zero by the 2000s. Interestingly, the long-run return to autarky for capital owners in China (the black line) declined from the early 1990s through 2005, and then recovered, mirroring the evolution of the ACR component in Figure 1.

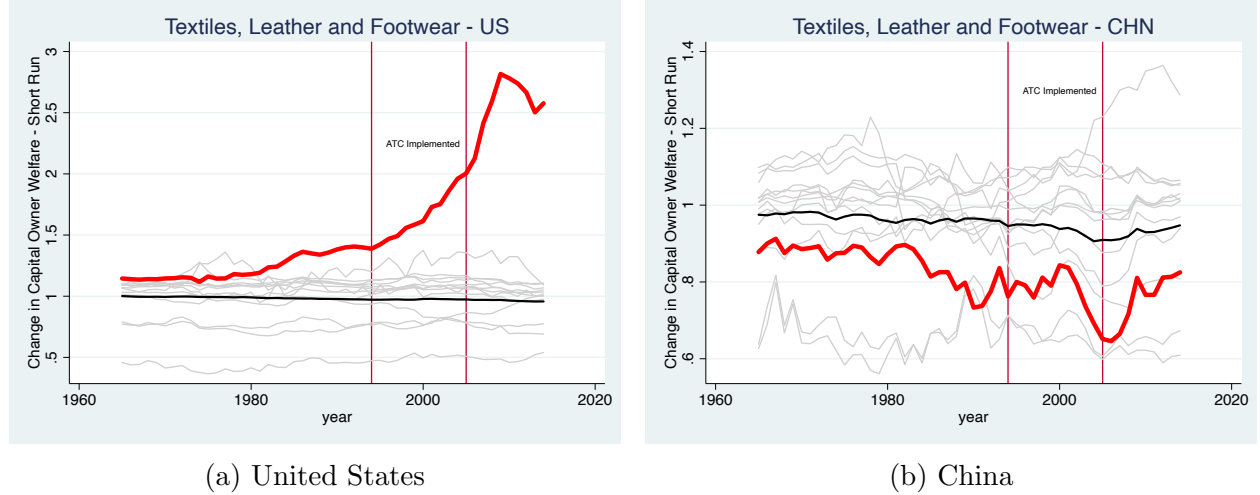


Figure 2: Short-Run Returns to Autarky for Capital Owners in Textiles, Leather and Footwear

Notes: The thick red line plots $W_{cap,ij}^a/W_{cap,ij}$ for textiles, leather and footwear for a return to autarky from each year. Gray lines are other industries, and the black line is the long-run change in real income for capital owners, $W_{cap,j}^a/W_{cap,j}$. The thin vertical line marks the implementation of the Agreement on Textiles and Clothing, and the thick vertical line marks the removal of the remaining quotas in 2005.

8 Conclusion and Next Steps

In this paper, we develop an adjusted gains from trade formula that captures the presence of specific factors, and show that the cost of capital being stuck in its industry can be quantified using national accounts data alone. Our results suggest minimal aggregate consequences of specific factors in the US, and larger costs in China and in a number of small, open economies. However, the returns to individual capital owners can vary substantially from the short to the long run, and some industries, as capital owners, have changed drastically in terms of their willingness to accept autarky over the last half-century.

Our analysis compares two polar cases: capital that is entirely fixed within industries, and capital that is perfectly mobile. In practice, the transition from the short to the long run depends on the fungibility of capital across industries. For example, agricultural capital is likely trapped within agriculture, and would depreciate significantly if moved to another industry, as tractors cannot easily be used by a technology firm. In contrast, capital embodied in some manufacturing and service industries, such as computers, office space, and

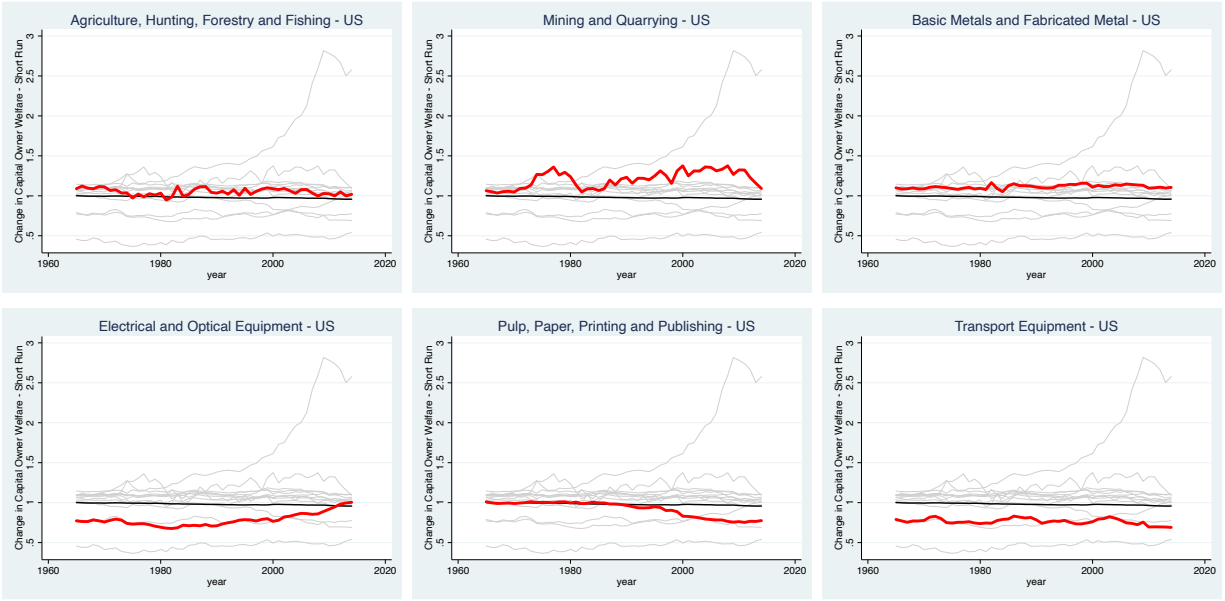


Figure 3: Short-Run Returns to Autarky for Capital Owners, Selected US Industries
Notes: See notes to Figure 2.

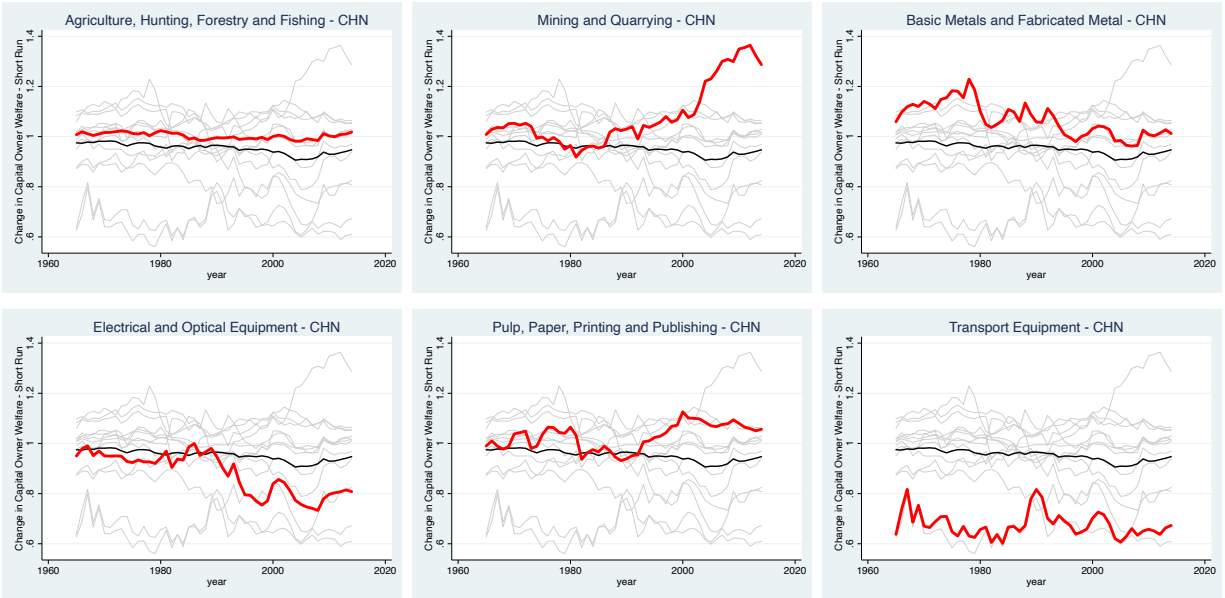


Figure 4: Short-Run Returns to Autarky for Capital Owners, Selected Chinese Industries
Notes: See notes to Figure 2.

vehicles, may be much more fungible. In ongoing work, we are developing a dynamic model of capital reallocation in which capital owners choose where to deploy their capital each period, and moving capital from one industry to another is subject to industry-pair-specific depreciation. Our hypothesis is that asymmetries in factor fungibility between industries will have qualitative impacts on the analysis of trade shocks, both for the speed of adjustment to the long run, and for which capital owners bear the costs along the transition path.

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A Exact Changes in Prices

To complete the derivation presented in Section 3, note that the industry i price index in j can be written as:

$$P_{ij}^{1-\sigma_{ij}} = \sum_{k \in J} \left(\tau_{kj} w_k^{\gamma_{ik}} P_k^{(1-\gamma_{ik})} a_{ik} \right)^{1-\sigma_{ij}} \quad (40)$$

Thus, the log change in the price index for industry i in country j is:

$$d \log P_{ij} = \sum_k \lambda_{ikj} (d \log \tau_{kj} + \gamma_{ik} d \log w_k + (1 - \gamma_{ik}) d \log P_k + d \log a_{ik}) \quad (41)$$

To simplify the interior of the sum, take the ratio of the trade share from k to j in i , and the share of j from itself in i :

$$\frac{\lambda_{ikj}}{\lambda_{ijj}} = \left(\frac{\tau_{kj} w_k^{\gamma_{ik}} P_k^{(1-\gamma_{ik})} a_{ik}}{\tau_{jj} w_j^{\gamma_{ij}} P_j^{(1-\gamma_{ij})} a_{ij}} \right)^{1-\sigma_{ij}} \quad (42)$$

Setting labor in j as the numeraire, and assuming that internal trade costs do not change, $d \log \tau_{jj} = 0$, log differentiating yields:

$$\begin{aligned} \frac{1}{1 - \sigma_{ij}} (d \log \lambda_{ikj} - d \log \lambda_{ijj}) &= d \log \tau_{kj} + \gamma_{ik} d \log w_k + (1 - \gamma_{ik}) d \log P_k + d \log a_{ik} \\ &\quad - (1 - \gamma_{ij}) d \log P_j - d \log a_{ij} \end{aligned} \quad (43)$$

Substituting into (41), and using $\sum_k \lambda_{ikj} = 1$ and $\sum_k d \lambda_{ikj} = 0$, we get:

$$d \log P_{ij} = \frac{1}{\sigma_{ij} - 1} d \log \lambda_{ijj} + (1 - \gamma_{ij}) d \log P_j + d \log a_{ij} \quad (44)$$

Integrating over a trade shock, and exponentiating, we get (17):

$$\frac{P'_{ij}}{P_{ij}} = \left(\frac{\lambda'_{ijj}}{\lambda_{ijj}} \right)^{\frac{1}{\sigma_{ij}-1}} \left(\frac{P'_j}{P_j} \right)^{1-\gamma_{ij}} \frac{a'_{ij}}{a_{ij}} \quad (45)$$

In the short run, since $(1 - \beta_{ij}) V_{ij} = C_{ij} (Q_{ij})$, solving for Q_{ij} and substituting into the equation for a_{ij} yields (15), and thus, with capital fixed and $w_j = 1$:

$$\frac{a'_{ij}}{a_{ij}} = \left(\frac{V'_{ij}}{V_{ij}} \right)^{\beta_{ij}} \left(\frac{P_j}{P'_j} \right)^{(1-\gamma_{ij})\beta_{ij}} \quad (46)$$

Substituting into the exact change in industry prices, we get:

$$\frac{P'_{ij}}{P_{ij}} = \left(\frac{\lambda'_{ijj}}{\lambda_{ijj}} \right)^{\frac{1}{\sigma_{ij}-1}} \left(\frac{P'_j}{P_j} \right)^{(1-\beta_{ij})(1-\gamma_{ij})} \left(\frac{V'_{ij}}{V_{ij}} \right)^{\beta_{ij}} \quad (47)$$

Finally, to aggregate to the exact change in the economy-wide price index, P_j , we substitute into the Cobb-Douglas aggregator, $P'_j/P_j = \prod_i (P'_{ij}/P_{ij})^{\theta_{ij}}$, to get:

$$\frac{P'_j}{P_j} = \left(\prod_i \left(\frac{\lambda'_{ijj}}{\lambda_{ijj}} \right)^{\frac{\theta_{ij}}{\sigma_{ij}-1}} \left(\frac{V'_{ij}}{V_{ij}} \right)^{\theta_{ij}\beta_{ij}} \right)^{\frac{1}{1-\sum_i \theta_{ij}(1-\beta_{ij})(1-\gamma_{ij})}} \quad (48)$$

The derivation in the long run is identical, replacing the exact change in a_{ij} with the following expression, from (16):

$$\frac{\tilde{a}'_{ij}}{a_{ij}} = \left(\frac{\tilde{r}'_j}{\tilde{r}_j} \right)^{\beta_{ij}} \left(\frac{P_j}{\tilde{P}'_j} \right)^{(1-\gamma_{ij})\beta_{ij}} \quad (49)$$

B Additional Results

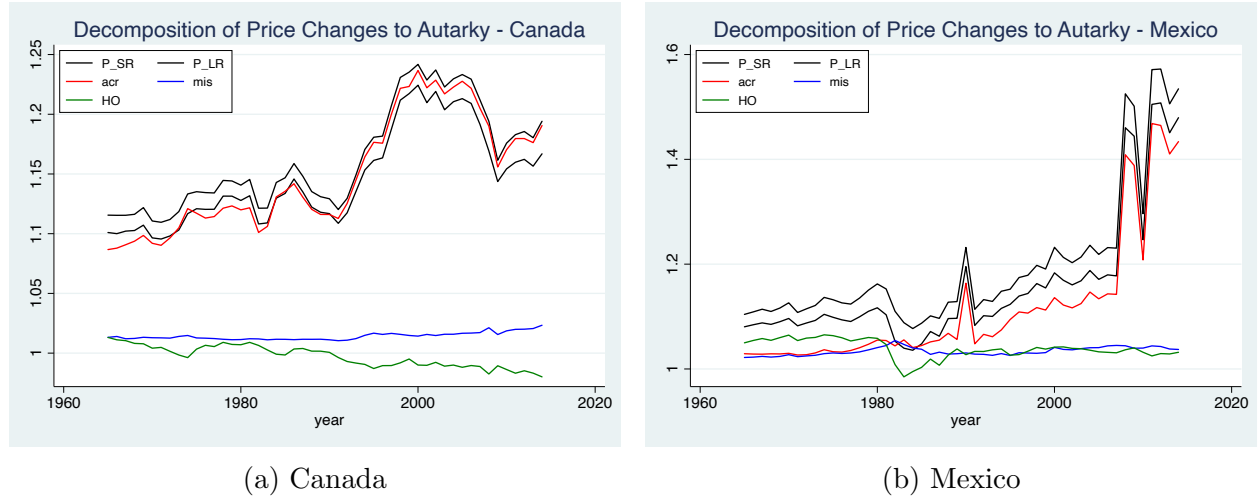


Figure 5: Decomposition of Price Changes from a Return to Autarky, 1965–2014: Canada and Mexico

Notes: See notes to Figure 1.

Table 3: Industry-Level Returns to Autarky for Capital Owners, Canada and Mexico, 2014

Industry	Canada		Mexico	
	$\frac{r_{ij}^a}{r_j}$	$\frac{W_{cap,ij}^a}{W_{cap,ij}}$	$\frac{r_{ij}^a}{r_j}$	$\frac{W_{cap,ij}^a}{W_{cap,ij}}$
Construction	0.230	0.192	0.096	0.062
Mining and Quarrying	0.544	0.456	0.610	0.397
Post and Telecommunications	0.912	0.764	1.196	0.779
Agriculture, Hunting, Forestry and Fishing	0.932	0.780	1.091	0.711
Basic Metals and Fabricated Metal	0.943	0.790	1.331	0.867
Wholesale and Retail Trade	0.996	0.834	1.011	0.659
Machinery, Nec	1.001	0.838	0.813	0.530
Transport and Storage	1.007	0.843	1.078	0.703
Manufacturing, Nec; Recycling	1.012	0.847	0.959	0.625
Hotels and Restaurants	1.054	0.882	1.200	0.782
Real Estate, Renting and Business Act.	1.066	0.892	1.238	0.807
Community Social and Personal Services	1.097	0.918	1.189	0.775
Financial Intermediation	1.132	0.948	1.342	0.875
Electricity, Gas and Water Supply	1.152	0.964	1.164	0.758
Pulp, Paper, Printing and Publishing	1.169	0.979	1.502	0.979
Coke, Refined Petroleum and Nuclear Fuel	1.172	0.981	1.303	0.849
Transport Equipment	1.254	1.050	0.800	0.521
Food, Beverages and Tobacco	1.335	1.118	1.204	0.785
Other Non-Metallic Mineral	1.423	1.192	1.128	0.735
Chemicals and Chemical Products	1.558	1.305	1.538	1.002
Rubber and Plastics	1.558	1.305	1.606	1.047
Electrical and Optical Equipment	2.153	1.803	0.847	0.552
Textiles, Leather and Footwear	4.604	3.856	1.306	0.851
Long run (all industries): $\tilde{r}_j^a/r_j, \tilde{W}_{cap}^a/W_{cap}$	0.951	0.815	1.049	0.709

Notes: Short-run results for a return to autarky starting from 2014. r_{ij}^a/r_j is the change in the rental rate in industry i relative to the wage under specific factors, from (33), and $W_{cap,ij}^a/W_{cap,ij}$ is the corresponding change in real income for owners of capital in industry i . Industries are sorted by $W_{cap,ij}^a/W_{cap,ij}$ for Canada. The last row reports the long-run change in the rental rate and real income for capital owners, which is common across industries.

Table 4: Short-Run Returns to Autarky for US Capital Owners over Time

Industry	1965	1975	1985	1995	2005	2014
Agriculture, Hunting, Forestry and Fishing	1.089	0.972	1.017	1.095	1.077	1.018
Mining and Quarrying	1.064	1.265	1.067	1.249	1.356	1.091
Food, Beverages and Tobacco	1.101	1.096	1.112	1.055	1.059	1.034
Textiles, Leather and Footwear	1.145	1.162	1.332	1.423	2.003	2.577
Pulp, Paper, Printing and Publishing	1.011	1.003	0.996	0.948	0.789	0.775
Coke, Refined Petroleum and Nuclear Fuel	1.096	1.203	1.144	1.100	1.169	1.002
Chemicals and Chemical Products	1.033	1.022	1.052	1.039	1.078	1.052
Rubber and Plastics	1.070	1.078	1.114	1.080	1.096	1.069
Other Non-Metallic Mineral	1.091	1.102	1.144	1.135	1.150	1.105
Basic Metals and Fabricated Metal	1.100	1.091	1.125	1.131	1.129	1.105
Machinery, Nec	0.458	0.366	0.466	0.476	0.509	0.540
Electrical and Optical Equipment	0.772	0.731	0.711	0.787	0.866	1.004
Transport Equipment	0.790	0.745	0.795	0.778	0.781	0.690
Manufacturing, Nec; Recycling	1.015	0.978	0.969	1.027	1.016	0.975
Electricity, Gas and Water Supply	1.042	1.087	1.101	1.081	1.187	1.166
Construction	0.138	0.169	0.148	0.175	0.147	0.207
Wholesale and Retail Trade	1.043	0.998	0.990	0.950	0.934	0.917
Hotels and Restaurants	1.122	1.091	1.094	1.079	1.053	1.046
Transport and Storage	1.013	0.986	0.985	0.927	0.939	0.915
Post and Telecommunications	0.884	0.910	0.924	0.967	0.825	0.809
Financial Intermediation	1.113	1.076	1.075	1.061	1.026	1.004
Real Estate, Renting and Business Act.	0.998	0.958	0.954	0.945	0.941	0.944
Community Social and Personal Services	1.110	1.078	1.081	1.058	1.042	1.034

Notes: Each cell reports $W_{cap,ij}^a/W_{cap,ij}$, the short-run change in real income for owners of capital in industry i from a return to autarky starting in the indicated year. Factor intensities are held fixed at their 2000–2014 medians.